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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BBA (HONOURS) REGULAR EXAMINATION MARCH 2026

FOURTH SEMESTER

Core Course (CC) - MG4CCRBBA202 - OPERATIONS MANAGEMENT

(2024 ADMISSION ONWARDS)

Part - A

Short Answer Questions (four to five sentences)

Each question carries 3 marks

1a. Identify two types of industries where Job Shop Production is adopted. [An] / [CO1]

Ans: Job shop production is adopted in industries where customized products are made based on specific customer requirements.

Examples (Any Two):

- Automobile repair workshop
- Tailoring/custom furniture
- Printing industry (custom printing jobs)
- Jewellery making industry (custom-designed ornaments)
- Machine tool manufacturing industry
- Interior design / carpentry workshops, etc

OR

1b. Identify two industries where batch production is commonly used. [An] / [CO1]

Ans: Batch production is commonly used in industries where products are manufactured in groups or batches**.**

Examples (Any Two):

- Food processing industry (biscuits, snacks)
- Cosmetics industry
- Chemical industry
- Footwear industry
- Paint industry
- Pharmaceutical industry, etc

2a. What is meant by the Five Ps of Production? [U] / [CO1]

Ans: The Five Ps of Production refer to the key elements that are essential for managing production activities effectively. The Five Ps of Production include:

- Product
- Plant
- Process
- People
- Programme

OR

2b. Identify any three core concepts involved in Operations Management. [U] / [CO1]

Ans: Any three from the following:

- Transformation Process
- Input–Output Relationship
- Value Addition
- Efficiency and Productivity

3a. Summarize two professional services and explain why they belong in this category. [U] / [CO2]

Ans: Professional services are knowledge-based, high-contact services**.**

Examples (Any Two):

Engineering Consultancy

- Provides technical solutions and project planning
- Requires specialized knowledge and expert skills

Management Consultancy

- Offers business strategies and organizational advice
- Involves problem-solving and client-specific solutions

IT Consulting Services

- Provides software, system and technology solutions
- Requires technical expertise and customization

Educational Services (Professors/Trainers)

- Impart knowledge and training
- Involves professional expertise and high interaction

Doctors (Healthcare Services)

- Provides diagnosis and treatment to patients
- Requires high customer interaction and customized care

Lawyers (Legal Services)

- Provide legal advice and represent clients in cases
- Involves specialized expertise and client-specific solutions

OR

3b. How does product volume influence process selection? [An] / [CO2]

Ans: Product volume plays a key role in selecting the type of production process.

- **Low Volume Production**

  → Job production is used

  → High flexibility and customization

- **Medium Volume Production**

  → Batch production is suitable

  → Moderate flexibility and variety

- **High Volume Production**

  → Mass or continuous production is adopted

  → High standardization and automation

As product volume increases, the production process becomes more standardized, automated and less flexible.

4a. Why is coordination difficult in fixed-position layout? [A] / [CO3]

Ans: Coordination becomes difficult because:

- Multiple teams work simultaneously at the same location
- Movement of resources leads to congestion and confusion
- Difficult to schedule activities efficiently
- Communication issues may arise among workers

OR

4b. What are the disadvantages of centralised purchasing? [A] / [CO3]

Ans: The following are the disadvantages:

- Delay in decision-making due to long procedures
- Lack of flexibility to meet local needs
- Ignores regional requirements and preferences
- Overburden on central authority
- Risk of inefficiency if communication is poor

5a. Differentiate between social sustainability and environmental sustainability. [An] / [CO4]

Ans:

Social Sustainability	Environmental Sustainability
Focuses on human welfare	Focuses on nature protection
Employee rights, safety	Resource conservation
Community development	Pollution control

OR

5b. Explain the importance of automation in modern industrial production. [An] / [CO4]

Ans: The Importance of Automation in Modern Industrial Production are as follows:

- Increases Productivity
- Improves Quality
- Reduces Labour Cost
- Saves Time
- Enhances Safety
- Ensures Consistency
- Supports Mass Production [5 x 3 = 15]

Part - B

Short Essay
Each question carries 5 marks

6a. Explain the functions of Operations Management in an organization. [K] / [CO1]

Ans: The functions of Operations Management are as follows:

- Selection of Product and Design
- Selection of Production Process
- Selecting Right Production Capacity
- Production Planning
- Production Control
- Quality and Cost Control
- Inventory Control
- Maintenance and Replacement of Machines

These functions ensure smooth production, efficient resource utilization and achievement of organizational goals**.**

OR

6b. Explain strategic, tactical and operational decisions with examples. [U] / [CO1]

Ans: Strategic Decisions: It is the long-term decisions taken by top management. They are broad, future-oriented and impact the entire organization

Examples:

- Selection of plant location
- Capacity planning
- Technology adoption

Tactical Decisions: It is the medium-term decisions taken by middle-level management. They **s**upport strategic decisions and focus on resource utilization

Examples:

- Production planning
- Inventory management
- Workforce scheduling

Operational Decisions: They are the **s**hort-term, day-to-day decisions taken by lower-level management**.** They are routine and repetitive.

Examples:

- Daily production scheduling
- Machine allocation
- Quality inspection

Type	Meaning	Example
Strategic	Long-term decisions	Plant location
Tactical	Medium-term decisions	Production planning
Operational	Short-term decisions	Daily scheduling

7a. Discuss the DMAIC methodology and explain the activities performed in each phase [U] / [CO4]

Ans: DMAIC is a data-driven problem-solving methodology used in Six Sigma to improve processes and eliminate defects.

Phases of DMAIC and Activities:

Define

- Identify the problem and project goals
- Understand customer requirements
- Prepare project charter

Measure

- Collect relevant data on current process performance
- Identify key performance indicators (KPIs)
- Establish baseline measurements

Analyze

- Identify root causes of problems
- Use tools like cause-and-effect diagrams
- Analyze data to find defects and inefficiencies

Improve

- Develop and implement solutions to eliminate root causes
- Test and validate improvements
- Optimize the process

Control

- Monitor the improved process
- Establish control systems and standards
- Ensure sustainability of improvements

OR

7b. Define Total Quality Management (TQM) and explain its objectives. [K] / [CO4]

Ans: Total Quality Management (TQM) is a management approach focused on continuous improvement of products, services and processes, involving all employees to achieve customer satisfaction and long-term success.

Objectives of TQM:

- To ensure customer satisfaction by meeting or exceeding customer expectations
- To achieve continuous improvement in products, services and processes
- To promote employee involvement and teamwork in quality management
- To enhance product and service quality by reducing defects and errors
- To reduce costs by minimizing waste and inefficiencies
- To improve process efficiency through standardization and control

8a. Discuss 5S audit and evaluation. [An] / [CO4]

Ans: 5S audit and evaluation refers to the systematic process of assessing the implementation and effectiveness of the 5S principles - Sort, Set in Order, Shine, Standardize and Sustain in an organization. It ensures that workplace organization practices are properly followed and continuously improved.

Importance of 5S Audit and Evaluation

- Ensures proper implementation of 5S practices
- Helps identify areas needing improvement
- Maintains discipline and standardization
- Improves productivity, safety and quality
- Supports continuous improvement (Kaizen)

OR

8b. Discuss the relationship between Industry 4.0 and smart manufacturing with suitable examples. [An] / [CO4]

Ans: Industry 4.0 refers to the integration of advanced digital technologies such as IoT, Artificial Intelligence, Big Data, robotics and cloud computing into industrial processes. Smart Manufacturing is the use of these technologies to create intelligent, automated and connected production systems.

Relationship between Industry 4.0 and Smart Manufacturing:

- Foundation and Application: Industry 4.0 provides the technological base, while smart manufacturing is its practical implementation in factories.
- Technology vs Usage: Industry 4.0 focuses on digital technologies, whereas smart manufacturing focuses on how these technologies are used in production.
- Connectivity and Integration: Both emphasize real-time data sharing, machine connectivity and system integration.
- Automation and Intelligence: Industry 4.0 enables smart manufacturing through automation, AI and self-decision-making systems.
- Efficiency and Flexibility: Together they improve productivity, flexibility and customization in manufacturing.

Examples of Industry 4.0: (Technology-focused)

- Internet of Things (IoT)
- Artificial Intelligence (AI)
- Big Data Analytics
- Cloud Computing
- Cyber-Physical Systems

Examples of Smart Manufacturing: (Application in production)

- Smart Factories
- Predictive Maintenance
- Automated Production Lines
- Real-time Production Monitoring
- Mass Customization [3 x 5 = 15]

Part - C

Long Essay

Each question carries 20 marks

9a. Evaluate the suitability of batch production for diverse industries, providing examples and justification. [E] / [CO2]

Ans: Batch production produces goods in groups or lots. Suitable for industries requiring balance between variety and volume.

Suitability:

- Medium scale industries
- Variety of products
- Flexible production

Examples:

- Pharmaceuticals
- Bakery
- Garments

Advantages:

- Flexibility
- Better quality control
- Economies of scale

Disadvantages:

- Setup time
- Inventory costs

OR

9b. Production Planning and Control are interrelated and interdependent. Demonstrate. [A] / [CO2]

Ans: Production Planning and Control (PPC) involves planning production activities and ensuring their execution as per plan.

- **Production Planning** → Decides what, how, when and where to produce
- **Production Control** → Ensures that production is carried out according to the plan

Interrelationship between Planning and Control:

- **Planning Sets Targets – Control Ensures Achievement**
  - Planning defines goals and schedules
  - Control monitors whether these goals are achieved

- **Control Depends on Planning**
  - Without proper planning, control cannot function effectively
  - Control follows the standards set by planning
- **Feedback Mechanism**
  - Control provides feedback to planning
  - Helps in correcting deviations and improving future plans
- **Coordination of Activities**
  - Planning organizes resources
  - Control ensures proper coordination during execution
- **Continuous Process**
  - Planning and control work together continuously
  - One cannot exist effectively without the other

Example:

- A company plans to produce 1000 units per week (Planning)
- Actual production is monitored and corrected if there is a delay (Control)

10a. Explain in detail the concept of standard time and its components. [A] / [CO3]

Ans: Standard time is the time required by a trained worker to complete a task at a normal pace under standard working conditions, including necessary allowances. It is used for planning, scheduling, costing and performance measurement.

Components of Standard Time:

- **Basic Time (Normal Time)**
  - Time taken by a worker to complete a task at normal speed
  - Does not include any allowances
- **Allowances**

  Additional time added to basic time to cover interruptions and worker needs:
  - Relaxation Allowance: For rest and fatigue
  - Contingency Allowance: For small, unavoidable delays
  - Delay Allowance: For machine breakdowns or interruptions

Formula:

Standard Time = Basic Time + Allowances

OR

10b. Discuss the importance of work study in achieving higher productivity, cost reduction and efficient resource utilization. [A] / [CO3]

Ans: Work study is a systematic analysis of work methods and time to improve efficiency and productivity. It includes: Method Study (improving how work is done) and Work Measurement (determining time required)

Importance of Work Study:

- **Higher Productivity**
  - Eliminates unnecessary movements and delays
  - Improves work methods
  - Increases output with the same resources
- **Cost Reduction**
  - Reduces wastage of materials, time and labour
  - Minimizes idle time and inefficiencies
  - Lowers production cost
- **Efficient Resource Utilization**
  - Ensures optimal use of men, machines and materials
  - Avoids overuse or underutilization of resources
- **Improved Work Methods**
  - Identifies better and simpler ways of doing tasks
  - Standardizes best practices
- **Better Planning and Control**
  - Provides accurate time standards
  - Helps in scheduling and performance evaluation
- **Enhanced Working Conditions**
  - Reduces worker fatigue
  - Improves safety and job satisfaction

[2 x 20 = 40]

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